

Reference motion

reference state $\hat{s}_t$



Policy + simulator

current state s_t



Differential construction

$$\Delta_t = \varphi(\hat{s}_t) - \varphi(s_t)$$

Policy residual

$$\Delta^- = \Delta_t$$

Positive anchor

$$\Delta^+ = 0$$

Equalized discriminator

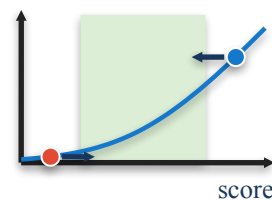


spectral normalization

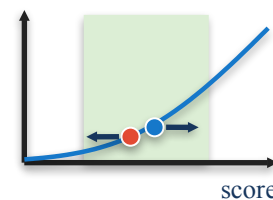
$$g_\theta(\Delta_t)$$

Anchor calibration

gap too large



gap too small



after calibration



- response band
- policy residual
- anchor

$$f_\theta(\Delta_t)$$

$$r^{DARE} = \text{softplus}(f_\theta(\Delta_t))$$

update policy π